

CHAPTER-1

INTRODUCTION

Coronary Artery Disease is the leading cause of death globally, accounting for approximately 17.9 million cardiovascular deaths annually [1]. It is caused by the progressive narrowing of the coronary arteries due to atherosclerotic plaque accumulation, resulting in impaired blood flow and increased risk of myocardial infarction and related complications. Early identification of at-risk individuals before symptom onset is central to reducing disease burden and enabling preventive interventions [2].

Risk prediction models play a critical role in this process by stratifying patients based on disease likelihood and guiding targeted interventions [3]. However, widely used models rely primarily on demographic, lifestyle, and clinical variables such as age, blood pressure, cholesterol, and smoking status [2], while largely excluding genetic predisposition that is known to explain around 40-60% of variation in susceptibility [4][5][6]. This limitation can lead to underestimation of risk, particularly in individuals without overt clinical symptoms [7].

India represents a high-risk population, contributing nearly one-fifth of global cardiovascular mortality, with CAD presenting at earlier ages and greater severity compared to Western populations [8]. These differences, combined with growing evidence of inherited susceptibility, highlight the need for population-specific risk prediction approaches. Conventional tools such as the Framingham Risk Score, developed primarily on Western cohorts, have been shown to underestimate CAD risk in South Asian populations [9].

Polygenic risk scores (PRS), which aggregate the effects of multiple genetic variants, provide a more direct and quantitative measure of inherited risk and have demonstrated improved identification of high-risk individuals beyond traditional factors, particularly in younger and clinically asymptomatic populations [2][3][8]. In parallel, digital twin technology enables the creation of dynamic, patient-specific models that integrate longitudinal clinical and physiological data for real-time monitoring and risk prediction [10][11][12]. Despite their potential, both approaches remain largely siloed, with limited clinical deployment and a lack of integrated frameworks combining genomic and dynamic health data [11][13][14].

To address these limitations, this work proposes a hybrid multi-stage framework that integrates a population-level PRS with lifestyle and clinical models to support cardiovascular risk prediction. The proposed system includes:

1. Computation of an India-specific population-level PRS using PGS2809 weights and GenomeIndia allele frequency data
2. Training of separate lifestyle and clinical risk models on large-scale datasets
3. Integration of genetic risk as a calibrated baseline offset within both models
4. Independent calibration of model outputs using a population-level genetic risk adjustment
5. SHAP-based explainability to quantify feature and domain-level contributions.

In addition, a machine learning-based counterfactual intervention engine is used to estimate potential changes in risk under modifiable lifestyle scenarios, forming the core digital twin component, with physiological simulation for haemodynamic validation. This framework aims to provide a more interpretable and population-aware approach to CAD risk assessment while enabling individualized what-if analysis for preventive decision support.

CHAPTER-2

LITERATURE SURVEY

Cardiovascular disease (CVD) remains the leading cause of global mortality, with a disproportionately high burden in South Asian populations. This project addresses the gap in India-specific, genetically informed CAD risk prediction by integrating polygenic risk scores (PRS), clinical data, and lifestyle factors within a digital twin framework. The following six studies directly motivate and validate the core components of this pipeline.

2.1 Roth, G.A., et al. (2020) - Global Burden of Cardiovascular Diseases and Risk Factors, 1990–2019. *Journal of the American College of Cardiology*, 76(25), 2982–3021.

This landmark study establishes CVD as the foremost preventable cause of global mortality, driven by aging populations and modifiable risk factors including hypertension, dyslipidemia, and obesity. It demonstrates that a substantial proportion of CVD burden can be mitigated through early, personalized risk stratification. For this project, it provides the epidemiological foundation for building a multimodal risk prediction system that integrates clinical, lifestyle, and genetic dimensions rather than relying on single-axis assessments.

2.2 Inouye, M., et al. (2018) - Genomic Risk Prediction of Coronary Artery Disease in 480,000 Adults. *Nature Medicine*.

This study demonstrates that a genome-wide PRS can identify individuals at elevated CAD risk decades before clinical symptom onset. It validates the use of PRS as an independent early-warning signal rather than merely a supplementary metric. This directly supports our approach of computing a population-level expected PRS using PGS2809 weights combined with GenomeIndia allele frequencies, enabling India-calibrated genetic risk estimation even in the absence of individual genotype data.

2.3 Sun, L., et al. (2021) - Polygenic Risk Scores in Cardiovascular Risk Prediction. *PLoS Medicine*, 18(1), e1003498.

This study evaluates the incremental predictive value of PRS when layered on top of traditional clinical risk factors, showing improved risk discrimination, particularly for individuals at intermediate clinical risk the segment most affected by misclassification. It supports the core design decision in this project to treat PRS not as a standalone predictor but as an additional feature domain alongside clinical and lifestyle inputs, with SHAP-based domain attribution used to quantify each domain's contribution to individual risk scores.

2.4 Martin, A.R., et al. (2019) - Clinical Use of Current Polygenic Risk Scores May Exacerbate Health Disparities. *Nature Genetics*, 21(5), 584–590.

This paper systematically demonstrates that PRS models trained predominantly on European GWAS cohorts exhibit significantly degraded performance when applied to non-European populations, with prediction accuracy declining sharply for South Asian and African ancestries. This directly motivates the

India-calibration step in our pipeline, where PGS2809 SNP weights are re-anchored to Indian-ancestry allele frequencies from the GenomeIndia dataset (9,768 individuals), and uncertainty in per-SNP frequencies is propagated through Monte Carlo simulation to produce 95% confidence intervals on the PRS estimate.

2.5 Prabhakaran, D., et al. (2016) - Cardiovascular Diseases in India: Current Epidemiology and Future Directions. *Circulation Research*, 118(11), 1874–1887.

This study characterizes the unique epidemiological profile of CVD in India, including earlier age of onset, higher case fatality rates, and risk factor distributions that differ substantially from Western cohorts and highlights the near-complete absence of validated India-specific risk prediction tools. It establishes the clinical necessity for the population-aware calibration approach taken in this project and explicitly justifies the limitation that our clinical training data (Cleveland and Hungarian cohorts) is derived from Western populations, a gap that the GenomeIndia-calibrated PRS component partially addresses.

2.6 Björnsson, B., et al. (2022) - Digital Twins to Personalize Medicine. *npj Digital Medicine*, 5(1), 4.

This study presents the digital twin framework as a mechanism for integrating heterogeneous data modalities, genomic, clinical, and physiological - into a unified, updateable patient representation capable of supporting intervention simulation and longitudinal risk projection. It establishes that a machine learning-based digital twin, without necessarily requiring organ-level physiological simulation, is both clinically meaningful and computationally tractable. This directly supports the architecture of our system, which uses a counterfactual intervention engine to simulate the effect of modifiable risk factor changes (e.g., weight reduction, physical activity) on predicted 10-year CAD risk, with outputs presented as ranked intervention plans with quantified risk reductions.

CHAPTER-3

ARCHITECTURE WITH EXPLANATION

3.1 System Overview

The system is implemented as a modular hybrid multi-stage pipeline in which each stage produces outputs used by subsequent stages. The pipeline processes three independent data streams: a lifestyle dataset (70,000 records), a clinical dataset (1,190 records), and genetic reference data. These inputs are preprocessed separately and then linked through a genetic calibration stage where a population-level polygenic risk score (PRS) is computed.

The pipeline then splits into two parallel branches for model training (lifestyle and clinical), which are processed as independent modelling pipelines. Each branch undergoes PRS-based risk adjustment followed by probability calibration.

Since the two datasets correspond to different cohorts and do not share patient-level alignment, their outputs are not combined through stacking or fusion. Instead, they are used as separate but complementary risk estimation pathways, depending on the availability of input features.

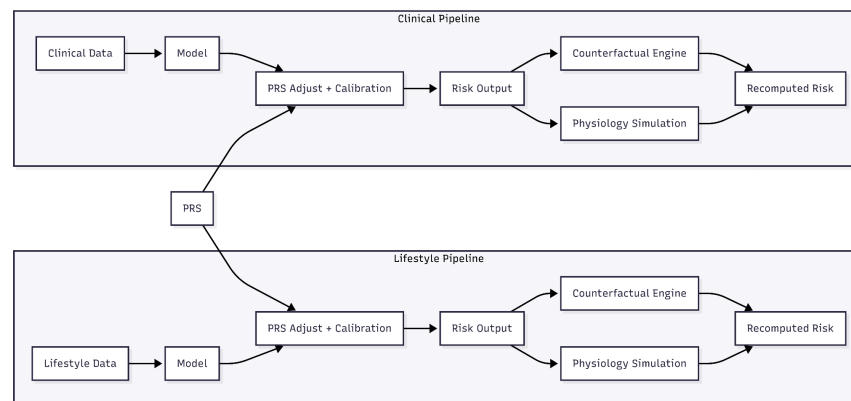


Figure 1 : Coronary Artery Disease (CAD) Risk Prediction Framework Using PRS, Lifestyle, and Clinical Data

3.2 Data Layer

The system uses three distinct data sources.

- The **lifestyle dataset** contains demographic and behavioural variables, including age, sex, blood pressure, BMI, smoking status, alcohol consumption, physical activity, and ordinal cholesterol and glucose categories. It is primarily used for large-scale model training.

- The **clinical dataset** contains continuous medical measurements such as cholesterol (mg/dL), resting blood pressure, maximum heart rate, ECG-related features, and angiographic indicators. This dataset provides higher clinical resolution and is also used for physiological simulation.
- The **genetic data** consists of PGS2809 SNP weights and GenomeIndia allele frequency data, which are used to compute a population-level PRS representing baseline genetic risk for the Indian population

3.3 Preprocessing and Feature Engineering

The lifestyle and clinical datasets are processed through separate pipelines due to differences in feature structure.

For the lifestyle dataset, Blood pressure values outside physiological ranges are removed, including cases where systolic pressure is less than or equal to diastolic pressure. BMI is computed from height and weight and added as a derived feature. Cholesterol and glucose are retained as ordinal variables. The dataset is split into training, validation, and test sets using stratified sampling (70/15/15), and feature scaling is performed using a StandardScaler fitted on the training data.

For the clinical dataset, preprocessing is performed on a unified combined dataset, with column standardization and handling of missing values through multivariate imputation using the IterativeImputer (MICE) approach. Sentinel values such as zero entries in features like cholesterol and resting blood pressure are first converted to missing values before imputation. Outlier handling is performed using interquartile range (IQR) based clipping, with bounds derived from the training set to prevent data leakage. The dataset is then split into stratified training and test sets, and an unscaled version of the test set is retained before preprocessing for downstream analysis and potential physiological simulation. Cross-validation is not applied at this stage but is instead used during the model training phase.

3.4 Genomic Processing and PRS Computation

SNPs from PGS2809 are matched to GenomeIndia using chromosome number and base pair position. For each matched SNP, allele alignment is performed to determine how the effect allele corresponds to the reference or alternate allele in GenomeIndia. If the effect allele matches the alternate allele, its frequency is used directly as the risk allele frequency. If it matches the reference allele, the complement of the alternate allele frequency is used. Palindromic SNPs (A/T and C/G) with allele frequencies between 0.42 and 0.58 are excluded to avoid strand ambiguity. SNPs without a positional match in the GenomeIndia dataset are also excluded.

The PRS is computed using the Hardy-Weinberg expected dosage approach. For each SNP, the expected contribution is calculated as twice the allele frequency multiplied by the effect weight. These contributions are summed across all retained SNPs to obtain a single population-level PRS value, representing the baseline genetic risk for coronary artery disease in the Indian population.

3.5 Machine Learning Model Training

Two independent machine learning models are trained on the lifestyle and clinical datasets. For each, six algorithms are evaluated: Logistic Regression, Random Forest, XGBoost or LightGBM, Support Vector

Machine, k-Nearest Neighbours, and a Multi-Layer Perceptron. Performance is assessed using five-fold stratified cross-validation, with final model selection based on AUC-ROC on a held-out stratified test set.

For the clinical dataset, additional hyperparameter tuning and regularization are applied to address overfitting due to the limited sample size. Model interpretability is incorporated using SHAP (TreeExplainer) for tree-based models, coefficient-based interpretation for linear models, and permutation importance for other models.

3.6 Genetic Risk Integration

The population-level PRS is incorporated as a post-model calibration component applied to both the lifestyle and clinical model outputs. The final predicted probability is computed as a weighted combination of the model prediction and a sigmoid-transformed PRS value, where a higher weight is assigned to the model output (0.85) and a smaller weight (0.15) is assigned to the PRS-based adjustment. This formulation reflects the role of PRS as a population-level genetic baseline offset rather than an individual-level predictor. The resulting calibrated probabilities are used to assign individuals into four risk categories: low, moderate, high, and very high based on predefined thresholds.

3.7 Independent Model Calibration

Since the lifestyle and clinical datasets consist of distinct, non-overlapping patient cohorts, traditional model stacking or feature fusion was not feasible. Therefore, the lifestyle and clinical pipelines operate as independent risk estimation pathways.

Within each pathway, machine learning predictions are first calibrated using Platt scaling to guarantee probability reliability. These cohort-specific probabilities are subsequently adjusted using the population-level PRS to account for the baseline genetic risk. This decoupled approach ensures robust, statistically valid predictions without introducing cross-dataset leakage or unsupported assumptions about patient overlap.

3.8 SHAP Explainability

SHAP is used to interpret model predictions by quantifying the contribution of individual features, with TreeExplainer applied to the tree-based model. Features are grouped into three domains: genetic, lifestyle, and clinical. For each patient, the absolute SHAP values of features within each domain are aggregated and normalised to provide a percentage breakdown of domain-level contributions. This enables the model output to be interpreted in terms of the relative influence of genetic, lifestyle, and clinical factors.

3.9 Hybrid Digital Twin

The digital twin component consists of two complementary layers. The primary layer is an ML-based counterfactual engine, following the approach described by [15]. It takes a patient's feature vector, applies predefined lifestyle changes (such as smoking status, increased physical activity, weight reduction, and reduced alcohol intake), and passes the modified inputs through the trained pipeline to estimate updated risk. The difference between the original and modified predictions is used to compute risk reduction.

The secondary layer integrates the Pulse Physiology Engine. Patient models are initialised using continuous

clinical features such as resting blood pressure, maximum heart rate, BMI, age, and sex. The system then simulates haemodynamic responses to interventions like exercise, weight loss, and smoking cessation. The resulting changes in blood pressure and heart rate are mapped back into the ML feature space, and the pipeline is re-run to obtain updated risk estimates. These Pulse-based estimates are compared with the ML-only counterfactual results. If both are in a similar range, this provides an additional physiological perspective supporting the ML-based predictions.

3.10 Validation and Output Layer

The final stage focuses on evaluating the performance of the system. Standard metrics such as AUC-ROC and Brier score are used to assess prediction accuracy and calibration, and results are compared with traditional methods like the Framingham risk score.

An ablation study is conducted during the evaluation stage to compare performance across different feature combinations and to assess how each component of the pipeline (genetic, lifestyle, and clinical) contributes to overall model performance.

A set of predefined clinical scenarios is also tested to ensure that the model behaves as expected for example, risk should decrease when smoking is removed or when physical activity is increased.

Finally, the risk reductions from the digital twin are compared with values reported in clinical studies for interventions such as smoking cessation, exercise, and weight loss. This ensures the results remain consistent with established clinical evidence.

CHAPTER-4

METHODOLOGY

4.1 Data Sources and Preprocessing

The system uses three data streams: lifestyle, clinical, and genetic data.

The lifestyle dataset (70,000 records) contains demographic and behavioural features, including age, sex, blood pressure, BMI, smoking, alcohol use, and physical activity, along with a binary cardiovascular disease target. Age is converted from days to years, BMI is derived, and blood pressure values outside physiological limits are removed. The dataset is split using stratified sampling (70/15/15), and StandardScaler is applied based on the training data. If class imbalance exceeds 1.5:1, SMOTE (Synthetic Minority Over-sampling Technique) is applied to the training set.

The clinical dataset combines Cleveland, Hungarian, and Statlog datasets (1,190 records) and includes continuous clinical variables such as cholesterol (mg/dL), resting blood pressure, maximum heart rate, and ECG-related features. Preprocessing includes data alignment, median imputation for missing values (ca, thal), removal of invalid records (e.g., cholesterol = 0), and outlier filtering.

An unscaled version of the test set is retained for physiological simulation. Since this dataset is derived from Western populations, it is used primarily for model validation, with population-specific adjustment handled through genetic calibration.

The genetic component uses PGS2809 and GenomeIndia. PGS2809 originally contains 205 SNPs, of which 182 are retained after alignment and filtering. GenomeIndia provides allele frequencies for Indian ancestry across approximately 129.9 million variants, enabling population-specific PRS computation

4.2 PRS Computation and Integration

SNPs from PGS2809 are matched to GenomeIndia using chromosome and position. Allele alignment is performed to determine whether the effect allele corresponds to the reference or alternate allele. If matched with the alternate allele, its frequency is used directly; otherwise, the complement is used. Palindromic SNPs with ambiguous frequencies are excluded, and unmatched SNPs are removed. Approximately 92% of SNPs are successfully matched. The PRS is computed using the Hardy–Weinberg expectation. For each SNP:

$$\text{PRS contribution} = 2 \times \text{allele frequency} \times \text{effect weight}$$

The total PRS is obtained by summing contributions across all retained SNPs. This produces a population-level genetic risk value representing expected CAD risk for an average Indian individual. This PRS is not used as an individual prediction but as a calibration factor. It adjusts baseline risk estimates to

account for population differences, similar in purpose to ethnicity adjustments but based on observed allele frequencies.

4.3 Machine Learning Model Training

Six machine learning models are evaluated on both datasets: Logistic Regression, Random Forest, XGBoost or LightGBM, Support Vector Machine, k-Nearest Neighbours, and a Multi-Layer Perceptron.

Five-fold stratified cross-validation is used for evaluation. For the clinical dataset, additional hyperparameter tuning is applied due to a smaller sample size, and overfitting is monitored using the train-test AUC gap. The best-performing model based on test AUC-ROC is selected for each dataset.

Class imbalance is handled using SMOTE on the training data when required. For the clinical dataset, class weighting is used where oversampling is not suitable

4.4 Risk Integration and Calibration

The PRS is incorporated into model predictions using a weighted combination:

$$p_{\text{final}} = 0.85 \times p_{\text{model}} + 0.15 \times \text{sigmoid}(\text{PRS})$$

where p_{model} is the probability of cardiovascular disease predicted by the machine learning model (lifestyle or clinical), and p_{final} is the PRS-adjusted risk probability. The sigmoid transformation maps the PRS to the probability scale.

This ensures that the machine learning model remains the primary contributor while the PRS provides a smaller calibration effect.

Since the lifestyle and clinical datasets correspond to different cohorts and do not share patient-level alignment, their predictions are not combined using stacking or fusion. Instead, each model is evaluated independently. To improve probability reliability, Platt scaling (sigmoid calibration) is applied separately to the outputs of each model using cross-validation.

4.5 Explainability and Digital Twin

Model predictions are interpreted using SHAP. Feature contributions are grouped into three domains: genetic, lifestyle, and clinical. For each patient, SHAP values are aggregated within each domain and normalised to provide a percentage breakdown of risk.

The digital twin is implemented as a counterfactual analysis framework. For each patient, modifiable features (smoking, physical activity, BMI, and alcohol use) are altered to simulate interventions. The modified inputs are passed through the pipeline to estimate updated risk. The difference from the original prediction gives the expected risk reduction.

Five intervention scenarios are evaluated, including smoking cessation, increased activity, weight reduction, alcohol reduction, and combined interventions.

An optional physiological layer using the Pulse Physiology Engine simulates changes in blood pressure and heart rate for selected interventions. These values are mapped back into the model inputs to obtain updated risk estimates.

4.6 Validation Strategy

Model performance is evaluated using AUC-ROC for discrimination and Brier score for calibration, with comparison to the Framingham risk score. An ablation study evaluates performance across different configurations (genetic, lifestyle, clinical, and combined).

The digital twin is validated by comparing predicted risk reductions with values reported in clinical studies and by testing predefined scenarios to ensure correct directional behaviour.

CHAPTER-5

IMPLEMENTATION

The implementation is organised as a sequence of functional modules corresponding to the stages of the pipeline. The preprocessing and model training components are fully implemented, while the later integration and fusion stages are structured and partially executed. All modules are developed in Python using pandas, NumPy, scikit-learn, and gradient boosting libraries.

5.1 Lifestyle Data Preprocessing

The lifestyle preprocessing module operates on the 70,000-record Cardio_Data.csv dataset. Blood pressure filtering removes records outside physiological limits (systolic <50 or >250 mmHg, diastolic <30 or >200 mmHg, or systolic $\leq$ diastolic), resulting in the removal of 1,247 invalid records.

BMI is computed from height and weight and added as a derived feature. Cholesterol and glucose are retained as ordinal variables. The dataset is split into training, validation, and test sets using stratified sampling (70/15/15).

No feature scaling is applied at this stage, as scaling is handled within the model training pipeline. Validation checks confirm the absence of missing values, correct data types, and consistent feature structure across splits.

5.2 Clinical Data Preprocessing

The clinical preprocessing module combines the Cleveland, Hungarian, and Statlog datasets into a unified dataset of 1,190 records. Column names are standardised to a common schema.

Missing values (*ca* and *thal*) are imputed using the median. Records with cholesterol equal to zero are removed, as this represents missing data in the original dataset. Basic filtering removes unrealistic physiological values.

The dataset is split using the same stratified approach. An unscaled version of the test set is retained for use in physiological simulation.

5.3 Genomic Data Processing

The genomic module aligns SNPs from PGS2809 with GenomeIndia allele frequency data. To manage memory usage, chromosome-wise loading is used, where only relevant chromosome files are processed. SNP matching is performed using chromosome and base pair position. Allele alignment determines whether the effect allele corresponds to the reference or alternate allele. Palindromic SNPs with ambiguous frequencies are excluded.

After matching and filtering, 182 SNPs are retained from the original 205. The output is a harmonised SNP map containing allele frequencies and effect weights

5.4 PRS Computation

The PRS is computed using the harmonised SNP map. For each SNP, the contribution is calculated as:

$$\text{PRS contribution} = 2 \times \text{allele frequency} \times \text{effect weight}$$

The total PRS is obtained by summing contributions across all retained SNPs. Basic validation checks are applied to ensure valid frequency ranges and the absence of missing values. A sensitivity check confirms that the PRS is stable under small variations in allele frequency.

5.5 Lifestyle Model Training

The lifestyle model training module evaluates multiple machine learning algorithms, including Logistic Regression, Random Forest, XGBoost, Support Vector Machine (linear variant), k-Nearest Neighbours, and a Multi-Layer Perceptron. All models are implemented using a unified pipeline structure, where feature scaling is handled internally using StandardScaler. This avoids the need for external scaling and prevents inconsistencies between training and inference. Probability calibration is applied using cross-validated sigmoid calibration to ensure reliable probability estimates. Model performance is evaluated using five-fold stratified cross-validation, and the best-performing model is selected based on AUC-ROC. The final model is saved as a single pipeline object that accepts raw input features and produces calibrated probability outputs.

5.6 Clinical Model Training

The clinical model training module follows the same pipeline-based structure as the lifestyle model but accounts for the smaller dataset size. Overfitting is monitored using the gap between training and test AUC, and hyperparameter tuning is applied to improve generalisation. Class imbalance is handled using class weighting where necessary.

The use of continuous clinical variables such as cholesterol, blood pressure, and heart rate allows the model to capture more detailed clinical patterns.

5.7 Genetic Integration

The genetic integration module incorporates the population-level PRS as a post-model calibration component applied to the outputs of both the lifestyle and clinical models. Rather than being used as an input feature during model training, the PRS is introduced as a fixed baseline adjustment reflecting population-level genetic risk. The integrated probability is computed as a weighted combination of the model prediction and a sigmoid-transformed PRS value, with weights chosen to prioritise model-derived risk while allowing a modest genetic calibration effect. This formulation aligns with the role of PRS as a population-level offset and is fully implemented as a standalone stage before the fusion module.

5.8 Calibration and Explainability

A stacked fusion module was initially considered to combine lifestyle and clinical model outputs. However, since the two datasets do not contain the same individuals, such a fusion approach would introduce invalid assumptions and potential data leakage.

Therefore, the final system evaluates the lifestyle and clinical models independently. Each model's output is first adjusted using PRS-based integration and then calibrated using Platt scaling to improve probability estimates.

Explainability is implemented using cohort-specific SHAP (SHapley Additive exPlanations) analyses. Feature contributions are scaled according to the model integration weights and aggregated alongside the deterministic population-level genetic calibration. This allows the decision-making process to be transparently grouped into discrete genetic, lifestyle, and clinical domains.

CHAPTER-6

RESULTS

6.1 Data Preprocessing

The lifestyle dataset initially contained 70,000 samples with no missing values. After removing physiologically implausible blood pressure readings (1.95%), 68,633 samples were retained. The dataset remained well-balanced ($\approx 50\%$ class distribution), eliminating the need for resampling. BMI was derived from height and weight, and categorical variables were encoded, resulting in a final feature set of 14 features.

The clinical dataset comprised 1,190 samples from merged sources. Sentinel zero values were treated as missing and imputed using MICE, with all preprocessing steps applied using training-set parameters to prevent data leakage. IQR-based clipping was used for outlier handling, and one-hot encoding expanded the feature space. After removing leakage-prone features, a clinically relevant subset of predictors was retained. Class balance remained stable across splits ($\sim 47\%/53\%$).

6.2 Genomic Preprocessing & PRS Computation

The PGS Catalog score PGS002809 (205 SNPs for CAD) was harmonized against the Genome India dataset (9,768 individuals; 129.9 million variants across 22 chromosomes) using a coordinate-based positional matching strategy with directional allele alignment. Of the 205 PGS SNPs, **182 (88.8%)** were successfully matched, and oriented no SNPs were lost due to allele mismatch. The remaining 23 were absent from the Indian reference panel at the queried positions.

Population-level PRS was computed using the standard dosage formula:

$$\text{PRS} = \sum (2 \times p_{\text{Indian}} \times \beta)$$

where p_{Indian} is the alternate allele frequency from the Genome India reference and β is the PGS effect weight. The computed raw PRS was **11.88** (95% CI: 11.87–11.89, Monte Carlo, 10,000 simulations). A sensitivity analysis under $\pm 10\%$ allele frequency perturbation showed a variation of 17.9%, confirming robustness ($< 20\%$ threshold). The top-10 SNPs contributed 23.0% of the total PRS; the single largest contributor was rs11591147 (chromosome 1, $\beta = 0.241$, contribution = 4.05%), a locus previously associated with LDL metabolism.

Metric	Value
PGS SNPs (input)	205
SNPs matched (Genome India)	182 (88.8%)
Population PRS (raw)	11.88

95% CI (Montecarlo, n=10,000)	[11.869, 11.894]
Sensitivity variation ($\pm 10\%$ AF)	17.9% — Robust
Top-10 SNP contribution	23.0% of total PRS

Table 1: Summary of Genomic Preprocessing and PRS Computation for Coronary Artery Disease (CAD) using PGS002809.

6.3 Model Training: Lifestyle Module

Four classifiers (Logistic Regression, Random Forest, XGBoost, SGD Classifier) were evaluated using 5-fold stratified cross-validation on the lifestyle training set. XGBoost achieved the highest cross-validation AUC of 0.799 ± 0.004 . All models were stable (AUC std < 0.03). Evaluation on the held-out test set confirmed XGBoost as the best performer.

Model	CV AUC	CV F1	Test AUC	Test Brier
Logistic Regression	0.790 ± 0.003	0.708	0.797	0.185
Random Forest	0.787 ± 0.004	0.714	0.798	0.183
XGBoost	0.799 ± 0.004	0.719	0.806	0.178
SGD Classifier	0.774 ± 0.004	0.689	0.792	0.188

Table 2: Performance comparison of lifestyle-based machine learning models using 5-fold cross-validation and test set evaluation for CAD risk prediction.

The selected XGBoost pipeline (wrapped in CalibratedClassifierCV) achieved a **test AUC of 0.806** and a Brier score of 0.178. Feature importance analysis identified systolic BP (36.4%), elevated cholesterol (31.7% combined across levels), and age (5.3%) as the dominant predictors consistent with established cardiovascular risk literature.

6.4 Model Training: Clinical Module

The same four-model framework was applied to the clinical dataset, augmented with Gradient Boosting as a fourth candidate and with hyperparameter tuning via GridSearchCV for the top-2 CV performers. XGBoost (tuned: max_depth=4, learning_rate=0.1, reg_lambda=2.0) and Gradient Boosting were the top two models by cross-validation AUC.

Model	CV AUC	Test AUC	Test Brier	Test F1
Logistic Regression	0.848 $\pm$ 0.033	0.853	0.157	0.779
Random Forest	0.868 $\pm$ 0.027	0.860	0.153	0.792
Gradient Boosting	0.879 $\pm$ 0.027	0.875	0.142	0.800
XGBoost (tuned)	0.880 $\pm$ 0.032	0.885	0.134	0.832

Table 3: Performance comparison of clinical-based machine learning models using 5-fold cross-validation and test set evaluation for CAD risk prediction.

The selected XGBoost clinical model achieved a **test AUC of 0.885** and a Brier score of 0.134 - meaningfully better calibrated than Logistic Regression (Brier = 0.157). On the held-out test set of 238 patients, the model yielded an accuracy of 82.4%, precision of 83.9%, and recall of 82.5%. The CV-to-test AUC gap was only 0.005, indicating no overfitting. The top predictors were sex (23.8%), ST depression (oldpeak, 17.9%), fasting blood sugar (12.0%), and maximum heart rate (8.7%).

6.5 Summary of Completed Pipeline Stages

Pipeline Stage	Key Output	Status
Lifestyle preprocessing (70k)	54,906 train / 13,727 test samples; 14 features	Complete
Clinical preprocessing (1,190)	952 train / 238 test samples; 10 features	Complete
Genomic harmonization (PGS002809 $\times$ Genome India)	182/205 SNPs matched	Complete
Population PRS computation	PRS = 11.88 (95% CI: 11.869–11.894)	Complete
Lifestyle risk model (XGBoost)	AUC = 0.806, Brier = 0.178	Complete
Clinical risk model (XGBoost)	AUC = 0.885, Brier = 0.134	Complete

Genomic risk integration	PRS-based probability adjustment applied	Complete
Digital Twin inference pipeline	—	Under development

Table 4: Summary of completed pipeline stages, key outputs, and implementation status in the CAD risk prediction framework.

The end-to-end Digital Twin inference pipeline remains under development and is not included in the present results.

6.6 Genetic Risk Integration and Calibration

The integration of the population-level PRS resulted in a uniform upward shift in predicted probabilities (mean increase ≈ 0.07), reflecting the higher baseline risk in the Indian population.

This integration did not affect model discrimination (AUC remained unchanged), but slightly worsened calibration (increase in Brier score), as expected due to the constant offset nature of the PRS.

Subsequent calibration using Platt scaling improved probability reliability:

- Lifestyle model: Brier score improved from 0.1861 to 0.1790
- Clinical model: Brier score improved from 0.1422 to 0.1357

6.7 Explainability and Domain Attribution

SHAP analysis identified clinically meaningful predictors across both models.

For the lifestyle model, systolic blood pressure, age, and cholesterol levels were the dominant contributors.

For the clinical model, ST depression (oldpeak), maximum heart rate, and cholesterol were the most influential features. Domain-level attribution showed that:

- Clinical features contributed the majority of the predictive signal ($\approx 82\text{--}98\%$)
- Lifestyle factors contributed a smaller but meaningful portion ($\approx 13\%$ in the lifestyle cohort)
- Genetic contribution was low ($\approx 1\text{--}4\%$), consistent with its role as a population-level calibration factor

These results confirm that genetic risk acts as a baseline adjustment rather than a primary driver of individual predictions.

CHAPTER-7

CONCLUSION AND FUTURE WORK

7.1 Summary of the System

This project presents a multi-source coronary artery disease (CAD) risk prediction system that integrates lifestyle, clinical, and genetic information into a unified framework. The system combines a 70,000-record lifestyle dataset, a 1,190-record clinical dataset, and genomic data from PGS2809 and GenomeIndia to produce risk predictions calibrated to the Indian population.

A key component of the system is the computation of a population-level polygenic risk score (PRS), derived by aligning PGS2809 variants with GenomeIndia allele frequencies and applying a Hardy–Weinberg-based formulation. This PRS is incorporated as a calibration factor to adjust model predictions for population-specific genetic risk.

The modelling pipeline consists of two independently trained machine learning models (lifestyle and clinical), which undergo rigorous meta-calibration and act as complementary risk estimators. Model predictions are further interpreted using SHAP-based explainability, providing a breakdown of risk contributions across genetic, lifestyle, and clinical domains.

A digital twin framework is designed to estimate the impact of lifestyle interventions using counterfactual analysis, with an optional physiological layer for simulation-based validation.

7.2 Key Contributions

- The system incorporates GenomeIndia allele frequencies to compute a population-level PRS, addressing the limitation of existing models that are primarily calibrated on European populations.
- The architecture uses parallel modelling pipelines for lifestyle and clinical data, enabling the use of heterogeneous datasets without requiring feature-level or patient-level alignment.
- SHAP-based domain grouping enables risk to be expressed in terms of genetic, lifestyle, and clinical contributions, improving the interpretability of model outputs.
- A counterfactual-based digital twin is designed to estimate risk changes under lifestyle interventions, providing a practical tool for risk analysis.

7.4 Future Work

The immediate next step in this work is the completion of the digital twin module, including full implementation and evaluation of the machine learning-based counterfactual intervention engine and integration of the Pulse Physiology Engine for haemodynamic simulation. This will enable estimation of intervention-specific risk reductions and provide a more comprehensive assessment of the system.

Beyond the current scope, several extensions can further strengthen the framework. Validation on Indian clinical cohorts is a priority, as it would allow direct assessment of model performance in the target population. Incorporating individual-level genetic data would enable computation of personalised PRS values, improving the precision of genetic risk estimation.

Integration of longitudinal and wearable data could support continuous updating of risk predictions, moving toward a dynamic monitoring system. Additionally, benchmarking against Indian-specific cardiovascular risk scores would improve clinical relevance. Finally, deployment in real-world settings would require the development of a user-facing interface along with compliance with regulatory and data protection requirements.

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